

A classical finite counterexample to the (Lip_p)-isometric state-span conjecture

Candidate solution packet for run `fa_banach_001`

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Status: candidate counterexample, likely valid. For every $p \geq 1$, the regular action of the classical group C_3 on a three-point path metric has $\overline{\text{span}}^{w^*} S_p(A) = A^*$, although $S_p(A)$ is a proper subset of the state space. This disproves Conjecture 4.13 of Chirvasitu.

1 Source conjecture

For a compact quantum group A acting on a compact metric space (X, d) , Chirvasitu defines a state $\psi \in S(A)$ to be (Lip_p)-isometric when

$$W_p(x \triangleleft \psi, y \triangleleft \psi) \leq d(x, y) \quad (x, y \in X).$$

The set of such states is denoted by $S_p(A)$. Conjecture 4.13 on PDF page 23 of arXiv:1407.0960 [1] states:

Let A act faithfully on the compact metric space (X, d) , and $p \geq 1$. The set of states in the weak*-closed linear span of $S_p(A)$ coincides with $S_p(A)$.

Definition 4.12 A state $\psi \in S(A)$ is (Lip_p)-isometric if

$$W_p(x \triangleleft \psi, y \triangleleft \psi) \leq d(x, y), \quad \forall x, y \in X. \quad (29)$$

We denote the set of (Lip_p)-isometric states by $S_p(A)$. ♦

It turns out that $S_p(A)$ is a compact convex sub-semigroup of $S(A)$, and hence its weak*-closed linear span is a subalgebra of A^* which we could use as the analogue of A_D^* in the proof of Theorem 4.4. However, there might be, in principle, non-(Lip_p)-isometric states in this subspace. It is tempting to conjecture that this cannot happen:

Conjecture 4.13 Let A act faithfully on the compact metric space (X, d) , and $p \geq 1$.

The set of states in the weak*-closed linear span of the set $S_p(A)$ from Definition 4.12 coincides with $S_p(A)$.

If this were true, the proof of Theorem 4.4 would go through and (Lip_p)-isometric envelopes would exist for all faithful C^* quantum actions.

References

Figure 1: Definition 4.12 and Conjecture 4.13 in the source (PDF page 23).

2 Idea of the proof

Classical states are probability measures on the acting group. For a transitive finite action, Haar measure sends every point of X to the same uniform distribution. Measures sufficiently close to Haar therefore send every pair of points to distributions whose Wasserstein distance is smaller than the least nonzero distance in X . Thus $S_p(A)$ contains a relative neighborhood of Haar and consequently spans the whole dual A^* .

This is incompatible with any non-isometric element of the group: its point mass is a state of A and hence lies in that full span, but it expands some distance and so is not in $S_p(A)$. The three-cycle acting on a three-point path is the smallest transparent example.

3 A finite transport estimate

We use $\text{TV}(\alpha, \beta) = \frac{1}{2}\|\alpha - \beta\|_1$ for probability measures on a finite set.

Lemma 1. *Let (X, d) be a finite metric space of diameter D . For all $\alpha, \beta \in \mathcal{P}(X)$ and $p \geq 1$,*

$$W_p(\alpha, \beta)^p \leq D^p \text{TV}(\alpha, \beta).$$

Proof. Put the common mass $\min\{\alpha(z), \beta(z)\}$ on (z, z) for every $z \in X$. The unmatched mass on either side has total mass $r = \text{TV}(\alpha, \beta)$. Couple those two residual measures arbitrarily. The diagonal part costs zero and every residual pair costs at most D^p , so the resulting transport plan has cost at most $D^p r$. $\square$

4 A general counterexample principle

Theorem 1. *Let a finite group G act faithfully and transitively on a finite metric space (X, d) . Suppose the action is not isometric. Put $A = C(G)$. Then, for every $p \geq 1$,*

$$\overline{\text{span}}^{w^*} S_p(A) = A^*, \quad S_p(A) \neq A^*.$$

In particular, the states in the weak-closed linear span of $S_p(A)$ do not coincide with $S_p(A)$.*

Proof. A state of $A = C(G)$ is a probability measure μ on G , and $x \triangleleft \mu$ is the pushforward of μ under $g \mapsto xg$. Let h be the uniform probability measure on G . By transitivity, $x \triangleleft h$ is the uniform probability measure u_X on X , independently of x .

Let $D = \text{diam } X$ and let $m = \min\{d(x, y) : x \neq y\} > 0$. Pushforward contracts total variation, hence for every $x, y \in X$,

$$\begin{aligned} \text{TV}(x \triangleleft \mu, y \triangleleft \mu) &\leq \text{TV}(x \triangleleft \mu, u_X) + \text{TV}(u_X, y \triangleleft \mu) \\ &\leq 2 \text{TV}(\mu, h). \end{aligned}$$

Lemma 1 gives

$$W_p(x \triangleleft \mu, y \triangleleft \mu)^p \leq 2D^p \text{TV}(\mu, h).$$

Thus every state satisfying $\text{TV}(\mu, h) < m^p/(2D^p)$ belongs to $S_p(A)$: the case $x = y$ is trivial, and otherwise the displayed Wasserstein distance is less than $m \leq d(x, y)$.

Consequently $S_p(A)$ contains a relative neighborhood of the interior point h in the probability simplex on G . Its real affine span is the whole mass-one hyperplane, and its real linear span is the space of all signed measures on G . After complexification, its complex linear span is A^* . Since A^* is finite-dimensional, weak* closure changes nothing.

It remains to see that $S_p(A)$ is proper. Since the action is not isometric, some group element expands a distance. Indeed, if an element strictly contracts some pair, its inverse expands the image pair; hence there are $g \in G$ and $x, y \in X$ with $d(xg, yg) > d(x, y)$. For the state δ_g ,

$$W_p(x \triangleleft \delta_g, y \triangleleft \delta_g) = W_p(\delta_{xg}, \delta_{yg}) = d(xg, yg) > d(x, y).$$

Therefore $\delta_g \notin S_p(A)$, although it lies in the linear span just proved to be all of A^* . $\square$

5 The explicit three-point example

Take $X = \{0, 1, 2\}$ with

$$d(0, 1) = d(1, 2) = 1, \quad d(0, 2) = 2,$$

and let $G = C_3 = \mathbb{Z}/3\mathbb{Z}$ act by $xg = x + g \pmod{3}$. This regular action is faithful and transitive. It is not isometric because addition by one sends the pair $(1, 2)$, at distance one, to $(2, 0)$, at distance two. Faithfulness in the source's coaction sense is also immediate: the regular action has trivial kernel (equivalently, its matrix coefficients contain all point masses in $C(C_3)$). Theorem 1 therefore disproves Conjecture 4.13 for every $p \geq 1$.

For completeness, the full-span claim can be witnessed by only three states. Let $h = (1/3, 1/3, 1/3)$ and choose $0 < \varepsilon < 2^{-(p+1)}$. Then

$$\mu_0 = h, \quad \mu_1 = h + \varepsilon(\delta_0 - \delta_2), \quad \mu_2 = h + \varepsilon(\delta_1 - \delta_2)$$

are states in $S_p(A)$, because $D = 2$, $m = 1$, and $\text{TV}(\mu_i, h) \leq \varepsilon$. As row vectors, their determinant is $\varepsilon^2 \neq 0$, so they span $A^* \cong \mathbb{C}^3$. Yet

$$W_p(1 \triangleleft \delta_1, 2 \triangleleft \delta_1) = W_p(\delta_2, \delta_0) = 2 > 1 = d(1, 2),$$

so $\delta_1 \notin S_p(A)$.

6 Verification and novelty status

The included script `code/verify_c3_transport.py` uses exact rational arithmetic and a finite min-cost-flow solver to compute all nine orbit-law transport problems for the three displayed spanning states at $p = 1, 2, 3, 4, 5, 8$. It verifies their determinant and the point-mass violation. The calculation is a regression check only; Lemma 1 and the proof above establish the result for every real $p \geq 1$.

A bounded novelty check on 17 August 2026 searched arXiv, exact web phrases, the visible citation record, and later same-author quantum-isometry papers for the arXiv id, paper title, “Conjecture 4.13,” “ $S_p(A)$,” “weak*-closed linear span,” and close variants involving (Lip_p) -isometric states. No later proof or counterexample to this span conjecture was found. Novelty is plausible, not certified.

7 Scope and human-review recommendation

This packet fully negates Conjecture 4.13, even for a classical finite group. It does not address the separate conjecture earlier in the source that the conjunction of all (Lip_p) conditions implies (D) -isometry. Human review should first check the total-variation neighborhood argument and the classical identification of states with group measures. All remaining claims reduce to the displayed three-point metric and elementary linear algebra.

References

- [1] A. Chirvasitu, *On quantum symmetries of compact metric spaces*, J. Geom. Phys. 87 (2015), 134–151; arXiv:1407.0960.